Morphic Studio Software Architecture Specification

Document 41 — Database Architecture & Data Modeling

Version: 1.0

Status: Approved

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1. Purpose

This document defines the canonical database architecture for Morphic Studio.

It specifies how data is structured, related, versioned, secured, and optimized to support an AI-native creative platform.

The database must ensure consistency, scalability, integrity, auditability, and long-term maintainability.

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2. Database Philosophy

The database follows these principles:

2.1 Single Source of Truth

Each category of information has one canonical location.

Examples:

- Story facts → Story Bible
- Creative intent → Creative Memory
- Media → Asset Library
- User identity → Users
- Project metadata → Projects

2.2 Immutable Identifiers

Every persistent entity receives a Globally Unique Resource ID (GUID) when created.

Examples:

Project

Character

Scene

Chapter

Asset

Location

Relationship

Timeline Event

Render Job

Export

User

GUIDs never change, even if names or metadata are modified.

2.3 Normalization with Practical Performance

Normalize relational data where appropriate while allowing carefully controlled denormalization for performance-critical features.

2.4 Referential Integrity

Foreign keys enforce valid relationships.

Deletion policies must be explicit:

- Cascade

- Restrict
- Soft Delete
- Archive

No orphaned records should exist.

2.5 Auditability

Important entities retain historical changes.

Version history should be available for:

Projects

Story Bible

Characters

World Data

Assets

Settings

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3. Primary Entity Groups

Core domains include:

Users

Projects

Story Bible

Creative Memory

Characters

Worlds

Locations

Lore

Assets

Production

Publishing

AI

Notifications

Analytics

Permissions

Audit Logs

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4. Core Tables

Users

Stores:

Identity

Authentication references

Preferences

Profile

Subscription

Activity metadata

Projects

Stores:

GUID

Owner

Title

Description

Language

Status

Visibility

Created

Updated

Archived

Story Bible

Stores canonical project knowledge.

Includes:

Characters

Locations

Lore

Rules

Timeline

Canon Events

Relationships

World Facts

Creative Memory

Stores:

Themes

Narrative Intent

Mood

Writing Style

Recurring Symbols

Creative Goals

AI Learning Context

Unlike the Story Bible, Creative Memory captures evolving creative direction rather than fixed facts.

Characters

Stores:

GUID

Name

Aliases

Biography

Appearance

Voice

Personality

Relationships

Character Arc

Current Status

Associated Assets

Worlds

Stores:

World Information

Governments

Cultures

Magic Systems

Technology Levels

History

Maps

Regions

Assets

Stores:

GUID

Type

Owner

Project

Storage Location

Metadata

Dimensions

Duration

Tags

Version

Generation Source

License

Checksum

Production

Stores:

Storyboard

Comic Pages

Animation Data

Voice Assignments

Music Tracks

Render Metadata

Timeline

Production Status

Publishing

Stores:

Exports

Publication History

Distribution Targets

Formats

Schedules

AI Tasks

Stores:

Prompt

Provider

Model

Status

Cost

Execution Time

Token Usage

Output References

Retry History

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5. Relationships

Examples include:

Project

↓

Story Bible

↓

Characters

↓

Relationships

↓

Scenes

↓

Assets

↓

Production

↓

Publishing

The schema should maintain referential integrity throughout.

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6. Versioning Strategy

Support:

Document versions

Asset versions

Story revisions

Character history

World history

Configuration history

Restoration should be possible without data loss.

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7. Search Strategy

Index searchable entities including:

Projects

Characters

Locations

Assets

Lore

Scenes

Dialogue

Notes

Search indexes should remain synchronized with primary data.

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8. Performance Strategy

Implement:

Indexes

Composite indexes

Pagination

Cursor-based navigation

Materialized views where appropriate

Caching

Read optimization

Batch processing

Performance should be monitored continuously.

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9. Security

Protect data through:

Encryption at rest

Encryption in transit

Role-based access

Audit logging

Soft deletion

Secure backups

Least-privilege database access

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10. Backup & Recovery

Support:

Automated backups

Point-in-time recovery

Version restoration

Disaster recovery

Backup validation

Recovery testing

Recovery objectives should be documented.

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11. Data Lifecycle

Every entity follows a lifecycle.

Create

↓

Update

↓

Version

↓

Archive

↓

Restore (optional)

↓

Permanent Deletion (administrative policies only)

Deletion should be deliberate and auditable.

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12. Acceptance Criteria

The database architecture is successful if:

- Canonical data has one source of truth.
- Relationships remain consistent.
- Queries scale efficiently.
- AI services retrieve reliable context.
- Version history is preserved.
- Future features integrate without schema redesign.
- Data integrity is maintained.

Conclusion

The Morphic Studio database architecture establishes a durable, scalable, and trustworthy foundation for storing every aspect of the creative ecosystem. By combining canonical data ownership, immutable identifiers, robust relationships, versioning, and performance optimization, the platform is prepared to support complex storytelling workflows while remaining maintainable as the product evolves.